

Have Average Weekly Earnings Kept Pace With UK Consumer Prices and Private Rents? An ONS Analysis 2015-2025

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Data Used:

All data was sourced from the Office For National Statistics 2025-2015 including:

- CPIH
- Average Weekly Earnings, Total Pay, 21 July 2026 data vintage
- Price Index of Private Rents
- Expenditure category CPIH data

It's important to note that Average Weekly Earnings covers Great Britain whilst CPIH and Private Index of Private Rents covers the UK. The comparison should therefore be interpreted as GB earnings vs UK prices comparison rather than a perfectly aligned UK estimate.

Introduction:

The UK experienced a substantial increase in consumer prices between 2015-2025, particularly during the inflationary episode of 2021–2023. However, higher prices alone cannot establish whether households become less able to afford consumption as affordability is dependent on income as well. Therefore this project examines the relationship between consumer prices, average weekly earnings and private rents between 2015 and 2025. Through examining CPIH inflation data measuring consumer price changes, I was also able to collate a summary of whether certain price categories of items have been less affordable.

Methodology:

Monthly ONS data was imported into Python using Pandas to filter and clean the data by their relevant series, convert dates and numerical fields into appropriate formats and restrict the analysis to 2015-2025. Monthly data was aggregated into annual averages smoothing intra-year volatility. CPIH was already expressed as an index with 2015 being the base index of 100, private rents and average weekly earnings were rebased to 2015=100 to enable direct comparison across measures with different units. Annual CPIH inflation and earnings growth was calculated from year to year changes in respective annual average indices, CPIH was also divided into six key categories. Additionally, a CPIH deflated earnings index was constructed $(\text{Earnings Index} / \text{CPIH}) \times 100$ to provide a measure of purchasing power relative to 2015. The cleaned CPIH, Earnings and Private rent data, were exported into csv files and merged into a master analytical data set using year as the common key. Finally, Matplotlib was then used to provide visualisations.

Results

Figure 1

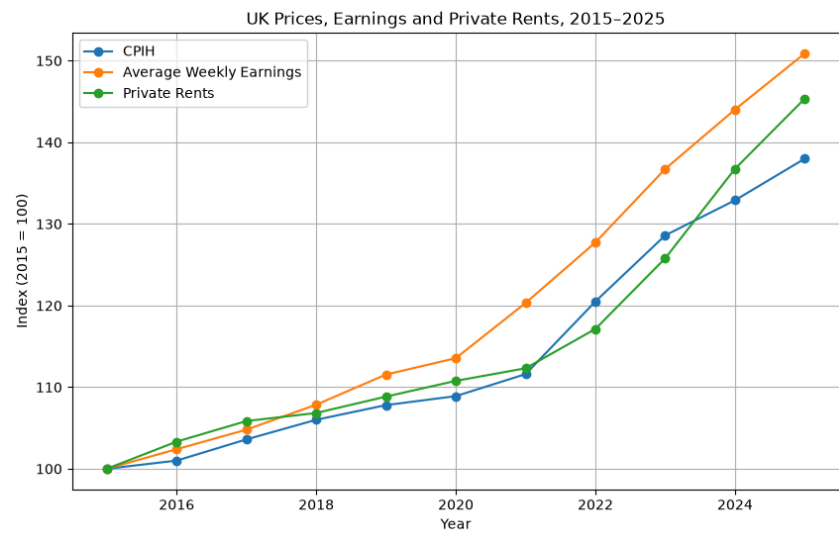
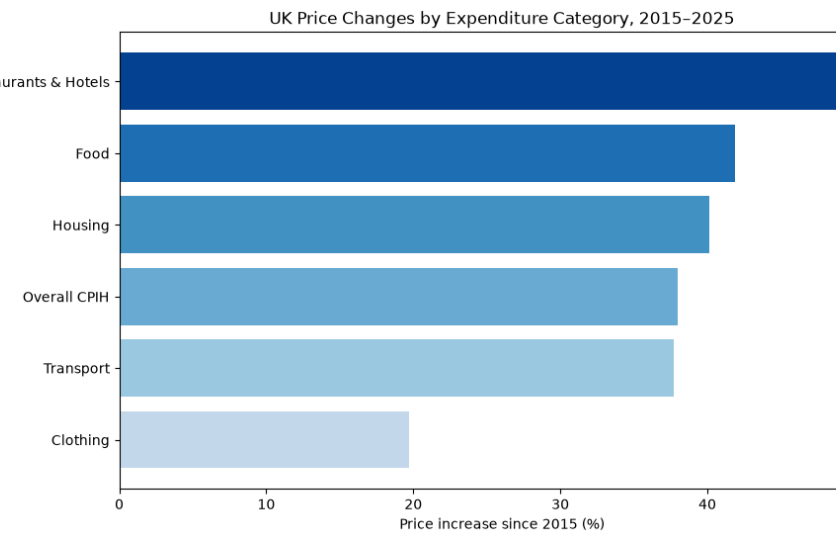


Table 1, Annual CPIH, Earnings and Private-Rent Indices, 2015–2025

Figure 2



Year	CPIH	Inflation	Cumulative_change	Weekly_Earnings	Earnings_Growth	Earnings_Index	Real_Earnings_Index	Rent_Index_2015	Rent_Change	Earnings_minus_Inflation
2015	100.00	-	-	£481.98	-	100.00	100.00	100.00	-	-
2016	101.00	1.00%	1.00	£493.53	2.40%	102.40	101.38	103.33	3.33%	1.40
2017	103.60	2.57%	3.60	£505.08	2.34%	104.79	101.15	105.85	5.85%	-0.23
2018	106.00	2.32%	6.00	£519.84	2.92%	107.85	101.75	106.84	6.84%	0.60
2019	107.80	1.70%	7.80	£537.63	3.42%	111.55	103.48	108.84	8.84%	1.72
2020	108.90	1.02%	8.90	£547.30	1.80%	113.55	104.27	110.77	10.77%	0.78
2021	111.60	2.48%	11.60	£579.91	5.96%	120.32	107.81	112.32	12.32%	3.48
2022	120.50	7.97%	20.50	£615.66	6.16%	127.74	106.01	117.13	17.13%	-1.81
2023	128.60	6.72%	28.60	£659.06	7.05%	136.74	106.33	125.79	25.79%	0.33
2024	132.90	3.34%	32.90	£694.16	5.33%	144.02	108.37	136.75	36.75%	1.99
2025	138.00	3.84%	38.00	£727.31	4.77%	150.90	109.35	145.36	45.36%	0.93

The data shows that the CPIH price index increased from 100 in 2015 to 138 in 2025, representing a 38% increase. Likewise, average weekly total earnings rose from an index of 100 to 150.9 whilst the private rent index increased to 145.4. Thus nominal earnings increased faster than both CPIH and Private Rent Index over the period.

Although, this relationship was not consistent every year as a reverse occurred in 2017 and 2022 with the real earnings index reaching 107.81 in 2021 then declining to 106.01 in 2022 and subsequently recovering to 109.35 in 2025.

Interpretation:

For most of the ten year interval, earnings growth exceeded inflation suggesting positive real earnings growth. Although nominal earnings increased by 50.9% whilst CPIH by 38%, a 12.9% difference, it is not itself a measure of the change in purchasing power. Instead I calculated a CPIH deflated real earnings index through dividing earnings by CPIH index, this index increased from 100 to 109.35 in 2025. This indicates an increase of approximately 9.35% in the purchasing power of average earnings relative to 2015. Yet it is important to note, that the earnings measure is seasonally adjusted for the whole economy including bonuses and may be affected by changes in workforce composition.

However for two years, inflation exceeded earnings growth: 2017 and 2022, having negative values in the earnings minus inflation column, -0.23 and -1.81. The negative gap in 2017 is consistent with research by Breinlich et al estimating that the depreciation of the pound value following the EU referendum increased UK inflation particularly for import exposed groups approximately by “1.7%” percent (Breinlich et al., 2017). This provides a possible explanation for the relatively weak earnings performance, although the data itself cannot identify causality, Whereas in 2022, Haskel, Martin and Brandt (2023) identified energy and food price shocks and labour market tightness displaced real wage increases, despite weekly earnings growing 6.16% that year (Haskel et al., 2023). Following 2022, the economy was able to recover with 0.33, 1.99, and 0.93, values in the three subsequent years.

2020 experienced low amounts of inflation, 1.02% coinciding with the stagnation of worldwide economies due to Covid-19 through weaker consumer demand and substantial falls in oil and transport prices (Sarwar, Abatti and Khalid, 2026). Similarly 2016 experienced the lowest inflation partly reflecting the earlier decline in global oil prices creating reduced energy and transport costs too.

It is important to consider how using average weekly earnings does have limitations as a measure of individual wage growth. It represents average employee earnings not accounting for the changes in the composition of the workforce or bonuses as the analysis uses total pay. The real earnings index should be instead interpreted as a measure of the purchasing power of average employee earnings rather than the change experienced by a typical worker.

Figure 2 illustrates a divergence across the selected CPIH expenditure divisions with food rising approximately 42% and CPIH housing, water, electricity, gas and other fuels division, 40%. Conversely, clothing experienced the slowest gain, 20% and transport approximately 37% reflecting the decade of volatile fuel costs and changing post pandemic commuting patterns. Finally, restaurants and hotels increased by nearly 50%, with the data identifying the magnitude of the price increase not the causes. These figures show that the price rises differed considerably between expenditure categories, although the chart measures price growth rather than the contribution of each category to overall inflation.

The UK price index of Private Rents increased by approximately 45.4% between 2015 and 2025 compared with a 50.9% increase in Great British average weekly earnings. This suggests that rents increased somewhat slower than the average earnings over the period. However, this data provides a general view of the entire country as there are substantial differences across locations, household sizes and incomes.

Conclusion

Overall, the given data suggests that Great Britain average earnings grew faster than CPIH between 2015 and 2025, However, this improvement was not uniform over time as real earnings declined in 2022 when CPIH inflation exceeded earnings growth. Furthermore, it does not conclusively prove that UK households were universally better off or increased by 9.35% as the earnings data cover Great Britain while CPIH and private rent data cover the UK. Living standards depend on factors beyond average wages and CPIH, including taxation affecting disposable income, household composition, distribution of income. The results therefore provide evidence of an improvement in purchasing power of average earnings rather than a complete measure of changes in living standards.

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